

Utilization of TDS Sensors for Water Quality Monitoring and Data Logging Using ESP32 and IoT

Abstract

This project implements a real-time TDS (Total Dissolved Solids) monitoring system using ESP32 and IoT-based technologies to assess water quality. The system uses a TDS sensor to detect the concentration of dissolved particles in water. The sensor is interfaced with an ESP32 microcontroller, which reads the TDS values and publishes them via MQTT to a central server. A Python-based backend logs these readings into a MySQL database, enabling historical analysis and future dashboard integration. This low-cost system enables remote, continuous monitoring of water turbidity levels essential for aquaculture, drinking water quality, and environmental management.

Keywords: TDS Sensor, ESP32, MQTT, IoT, MySQL, Water Quality, Data Logging, Python.

1. Introduction

Water quality is a critical parameter in both domestic and industrial applications. High levels of Total Dissolved Solids (TDS) indicate poor water quality and can be harmful to aquatic life and human health. Continuous monitoring using IoT-based solutions ensures early detection of contamination and automation of filtration systems.

The objective of this project is to build a smart TDS monitoring and data logging system. By utilizing ESP32 as the edge device, the system captures TDS values, sends them to a remote server via MQTT, and stores the data in MySQL using a Python script. The architecture is scalable and supports integration with cloud dashboards such as Grafana for real-time visualization.

2. Materials and Methods

2.1 ESP32

ESP32 is a dual-core microcontroller with Wi-Fi and Bluetooth capabilities. It is suitable for real-time IoT applications due to its low power consumption, robust wireless interface, and compatibility with the Arduino IDE.

2.2 TDS Sensor

The TDS sensor measures the concentration of dissolved solids in water, typically expressed in ppm (parts per million). The analog value is converted into a readable TDS level using calibration formulas. Standard ranges:(**Table 1**)

Table 1

Dissolved solid content

Content dissolved in water (mg/L)	Water assessment
<300 ppm	Excellent
300–600 ppm	Good
600–900 ppm	Acceptable
900 ppm	Poor to Hazardous

2.3 MQTT Protocol

MQTT (Message Queuing Telemetry Transport) is used for lightweight messaging between the ESP32 and the server. It enables efficient data transfer in real-time over the topic "watermonitor/tds".

2.4 MySQL and Python Backend

A Python script using Paho MQTT client and mysql-connector listens to TDS data and stores it in a MySQL table tds_log, with fields for tds_value and timestamp.

2.5 Work System Flowchart

The workflow design of the tool can be seen in **Fig. 1**.

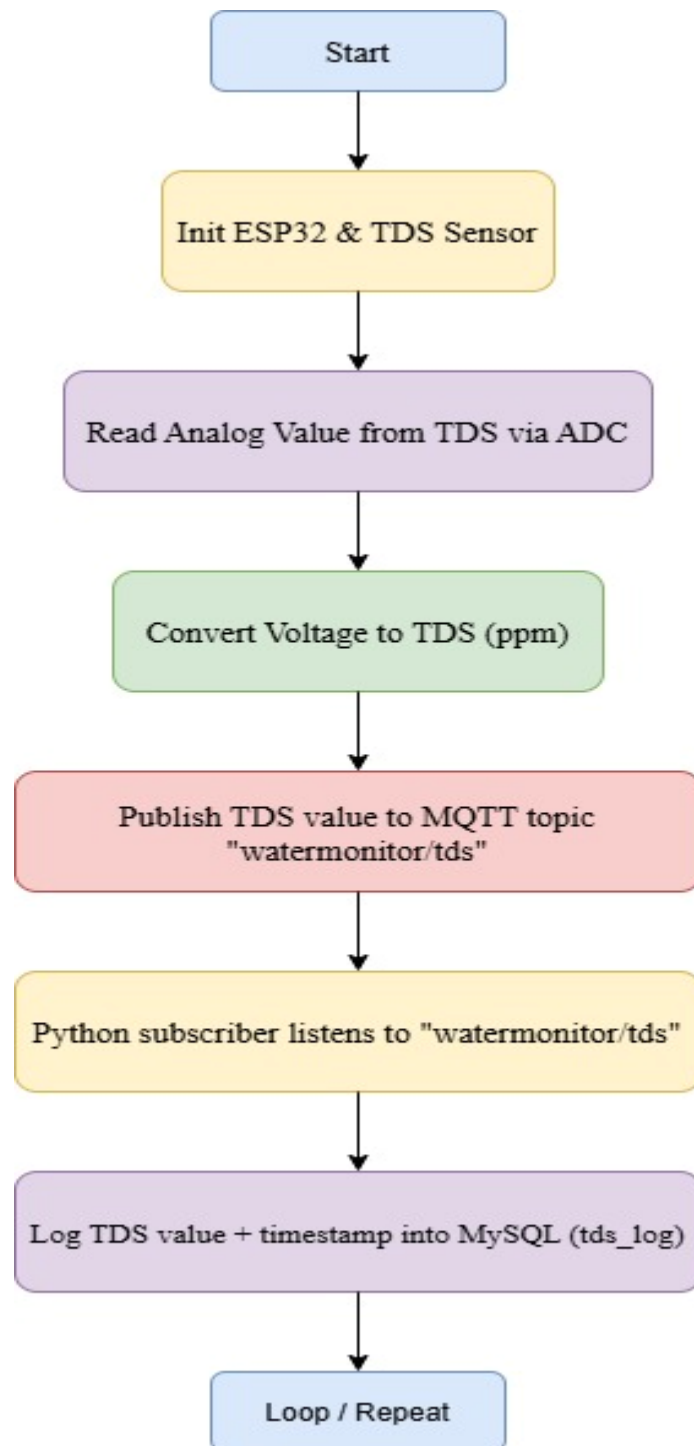


Fig 1 Work System Flowchart

2. 6. Overall Wiring Figure



Fig 2 Connection wiring

2. 7. The purpose of testing

The aim of this test is to determine the accuracy of the water TDS sensor reading by comparing it with a TDS measuring instrument. The sensor will be tested by taking clean water and dissolved pool water. This test will take samples 3 times. There are three types of water from which data will be collected, namely mineral water, water from pools that are still clean and pool water that is cloudy or dirty. In testing the TDS sensor, pool water data will be taken with a value of 0–600 Ppm.

This study will discuss data that has been taken in the field for analysis and data collection is carried out in a way that refers to the design of a water quality monitoring system and filtering catfish pond water based on turbidity based on the Internet of Things (IOT) with a TDS sensor and measuring directly with a tool. measure in the field. The purpose of testing and analyzing the design of a water quality monitoring system and filtering gold fish pond water based on turbidity based on the internet of things (IOT) with a TDS sensor is to determine the success of the program or tool that has been designed.

3. Software Implementation

3.1 Arduino Code (ESP32)

- Reads analog input from the TDS sensor.
- Converts raw value to TDS in ppm.
- Publishes value to MQTT broker every few seconds.

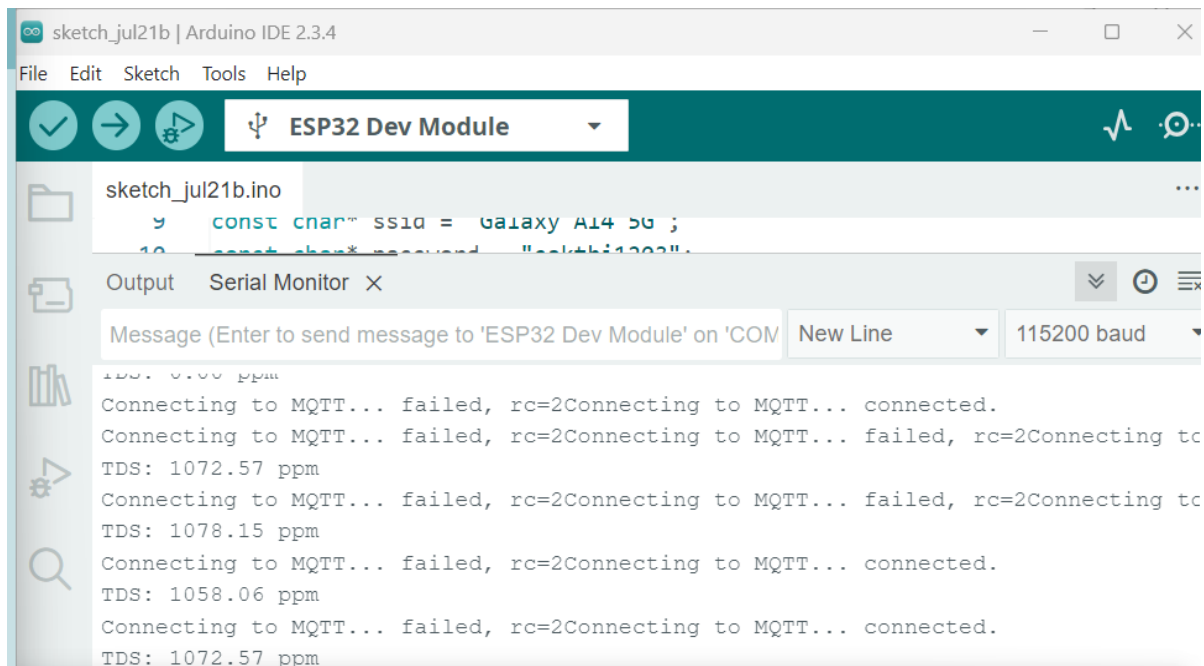
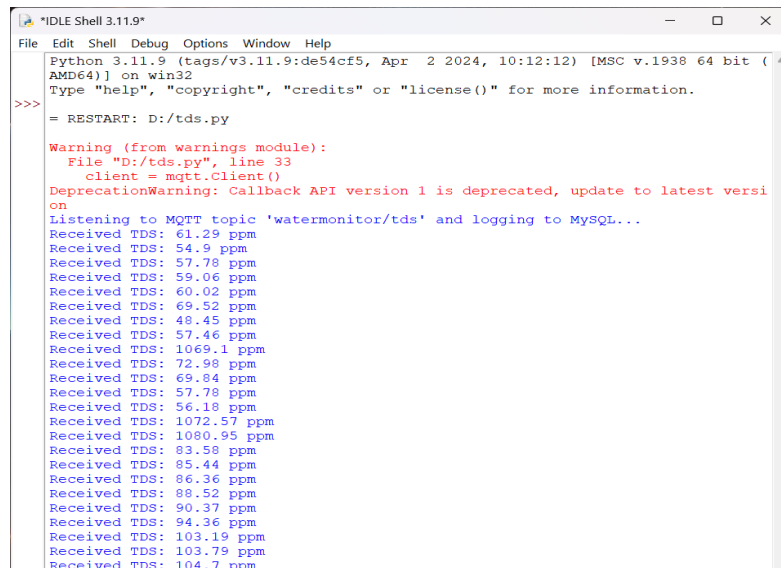


Fig 3 Arduino IDE Serial Monitor

3.2 Python MQTT Logger

- I. Subscribes to watermonitor/tds
- II. Parses incoming TDS values
- III. Logs them into a MySQL table with timestamp



```
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr 2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: D:/tds.py

Warning (from warnings module):
  File "D:/tds.py", line 33
    client = mqtt.Client()
DeprecationWarning: Callback API version 1 is deprecated, update to latest version
Listening to MQTT topic 'watermonitor/tds' and logging to MySQL...
Received TDS: 61.29 ppm
Received TDS: 54.9 ppm
Received TDS: 57.78 ppm
Received TDS: 59.06 ppm
Received TDS: 60.02 ppm
Received TDS: 69.52 ppm
Received TDS: 48.45 ppm
Received TDS: 57.46 ppm
Received TDS: 1069.1 ppm
Received TDS: 72.98 ppm
Received TDS: 69.84 ppm
Received TDS: 57.78 ppm
Received TDS: 56.18 ppm
Received TDS: 1072.57 ppm
Received TDS: 1080.95 ppm
Received TDS: 83.58 ppm
Received TDS: 85.44 ppm
Received TDS: 86.36 ppm
Received TDS: 88.52 ppm
Received TDS: 90.37 ppm
Received TDS: 94.36 ppm
Received TDS: 103.19 ppm
Received TDS: 103.79 ppm
Received TDS: 104.7 ppm
```

Fig 4 Python Serial Monitor

4. Results and discussion

4. 1. Purpose of Data Collection

The purpose of data collection is to find out the correctness of the work of the circuit and the operation of the water quality monitoring system and filtering of gold fish pond water based on turbidity based on the Internet of Things (IOT) with a TDS sensor. This test includes testing the turbidity sensor (TDS) and testing how the pool water filter works.

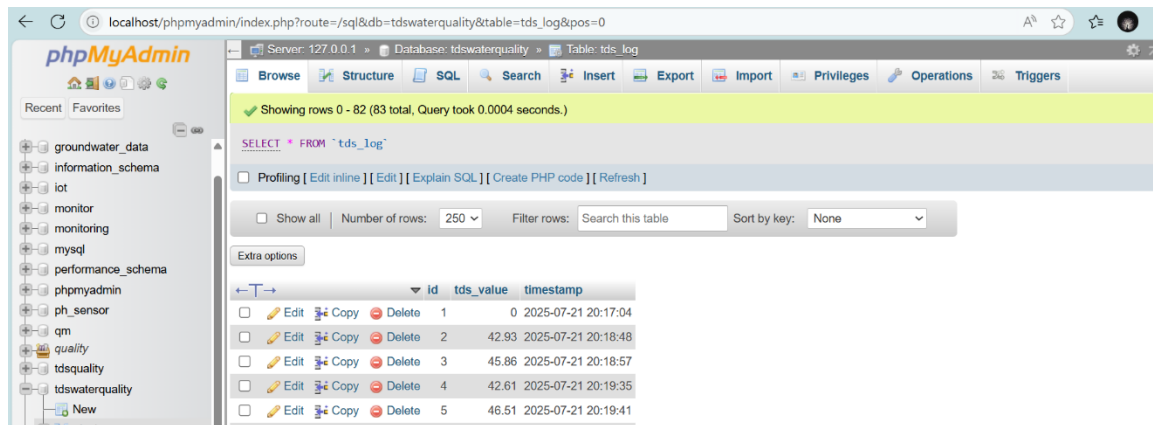
4. 2. Tools and Materials

S. No	Component	Description / Purpose
1	ESP32	Wi-Fi enabled microcontroller for reading sensor data and sending it via MQTT.
2	TDS Sensor	Measures the Total Dissolved Solids in water (in ppm).
3	Breadboard	Used for temporary circuit assembly during prototyping.
4	Jumper Wires	Connects the TDS sensor and ESP32 on the breadboard.
5	USB Cable	Powers the ESP32 and allows programming from a computer.

The purpose of this test is to find out the accuracy of the water TDS sensor reading by comparing it with the TDS measuring instrument. The sensor will be tested by taking clean water, tap water, Salt content water, this test will be sampled 3 times.

The results of test 1 are shown in **Fig 5**

Fig 5 TDS sensor trial results from clean water at 0–50 ppm.

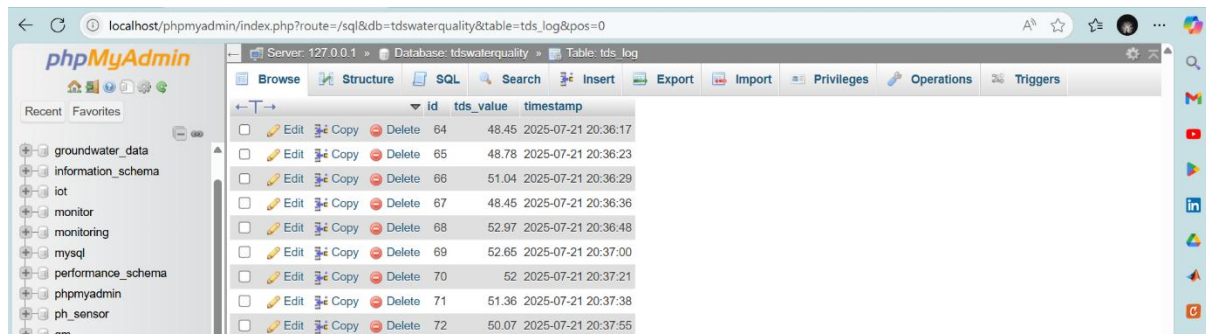


The screenshot shows the phpMyAdmin interface with the 'tds_log' table selected. The table contains 5 rows of data, showing TDS values ranging from 0 to 46.51 ppm. The interface includes a sidebar with a database structure tree, a top navigation bar, and a main content area with a SQL query editor and a table view.

id	tds_value	timestamp
1	0	2025-07-21 20:17:04
2	42.93	2025-07-21 20:18:48
3	45.86	2025-07-21 20:18:57
4	42.61	2025-07-21 20:19:35
5	46.51	2025-07-21 20:19:41

Fig 5 Database Table

Fig 6 TDS sensor trial results from tap water at 50–600 ppm.

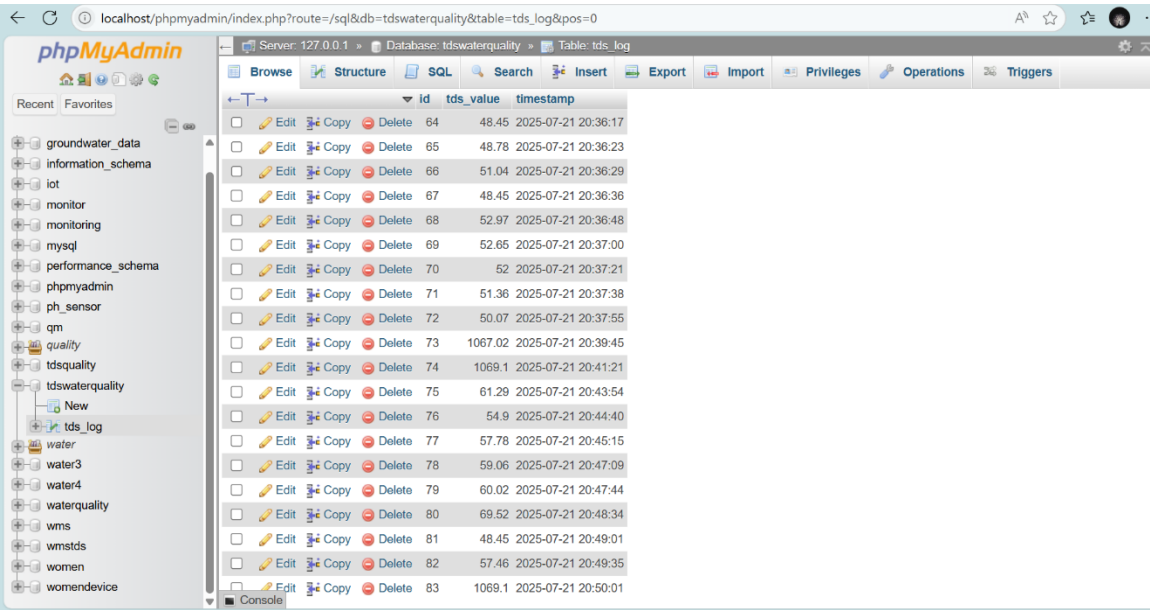


The screenshot shows the phpMyAdmin interface with the 'tds_log' table selected. The table contains 9 rows of data, showing TDS values ranging from 48.45 to 52.97 ppm. The interface includes a sidebar with a database structure tree, a top navigation bar, and a main content area with a SQL query editor and a table view.

id	tds_value	timestamp
64	48.45	2025-07-21 20:36:17
65	48.78	2025-07-21 20:36:23
66	51.04	2025-07-21 20:36:29
67	48.45	2025-07-21 20:36:36
68	52.97	2025-07-21 20:36:48
69	52.65	2025-07-21 20:37:00
70	52	2025-07-21 20:37:21
71	51.36	2025-07-21 20:37:38
72	50.07	2025-07-21 20:37:55

Fig 6 Database Table

Fig 7 TDS sensor trial results from Salt water at above 600 ppm.



id	tds_value	timestamp
64	48.45	2025-07-21 20:36:17
65	48.78	2025-07-21 20:36:23
66	51.04	2025-07-21 20:36:29
67	48.45	2025-07-21 20:36:36
68	52.97	2025-07-21 20:36:48
69	52.65	2025-07-21 20:37:00
70	52	2025-07-21 20:37:21
71	51.36	2025-07-21 20:37:38
72	50.07	2025-07-21 20:37:55
73	1067.02	2025-07-21 20:39:45
74	1069.1	2025-07-21 20:41:21
75	61.29	2025-07-21 20:43:54
76	54.9	2025-07-21 20:44:40
77	57.78	2025-07-21 20:45:15
78	59.06	2025-07-21 20:47:09
79	60.02	2025-07-21 20:47:44
80	69.52	2025-07-21 20:48:34
81	48.45	2025-07-21 20:49:01
82	57.46	2025-07-21 20:49:35
83	1069.1	2025-07-21 20:50:01

Fig 7 Database Table

4.3 Display of Water Quality

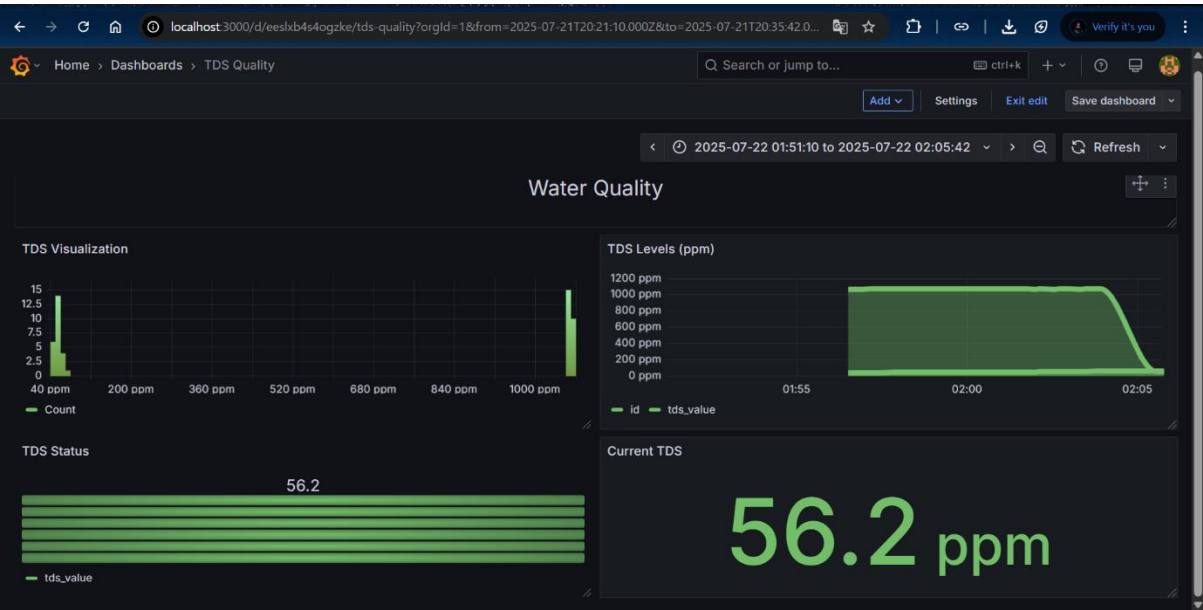


Fig. 8. Display of Water Trial Results 0 – 600 ppm

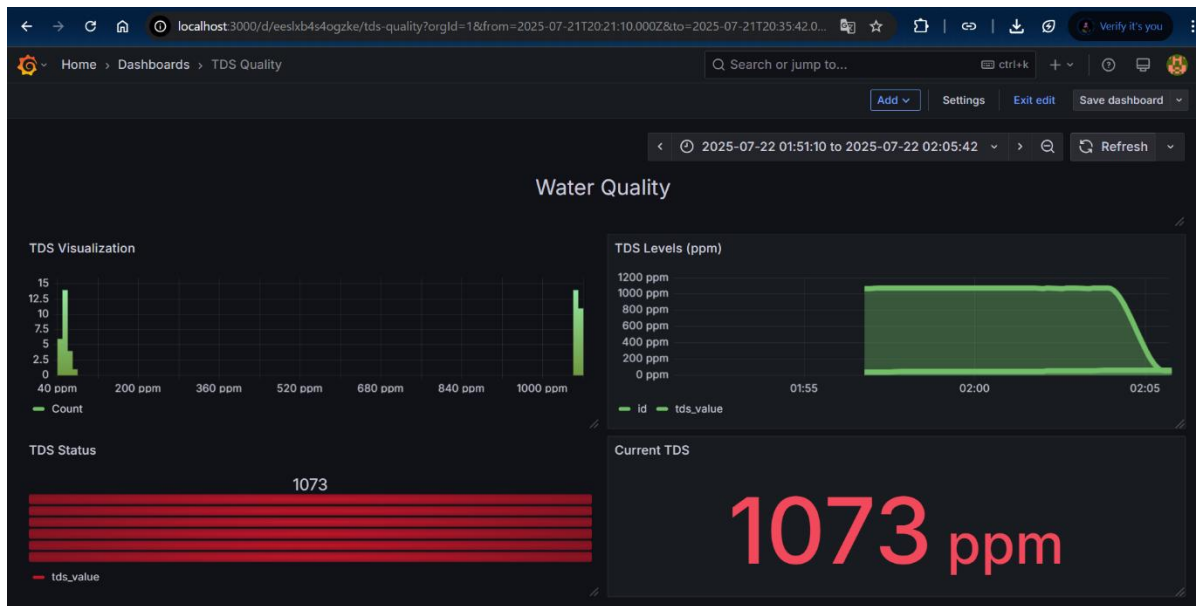


Fig. 9. Display of Water Trial Results above 600 ppm

5. Conclusion

The proposed IoT-based TDS monitoring system using ESP32, MQTT, and MySQL functions effectively. It accurately reads water quality data and ensures secure logging for future analysis. With simple enhancements, this system can support cloud dashboards, SMS/email alerts, or even automate filtration mechanisms.

6. Future Enhancements

- Add SMS alerts using Twilio
- Auto-trigger water purification system via relay module
- Web/Mobile dashboard for user interaction